

Information Security - Final - Spring 2025

Question 1:

Apply RSA algorithm to encrypt and decrypt after determining Public/Private PU/PR Keys. Moreover, determine n , $\phi(n)$ as well for the following parameters.

$p=7$, $q=17$, $e=11$, $m=11$

Question 2:

Apply ElGamal Digital Signatures to sign the message "World" using the parameters:

Prime number (p): 19, Primitive root (a): 13, Alice's Private Key (X_a): 6

Hash of message (m): 16, Alice chooses $K=5$

Question 3:

You are tasked to design Kerberos Authentication System. You have to explain the role of Authentication Server (AS) and Ticket Granting Server (TGS) so the higher authorities get a clear idea of a working of Kerberos Authentication.

Question 4:

Determine critically and prioritize the main cause of Denial of Service (DoS) attack in a large-scale online retail environment.

Question 5:

There are four different techniques used in Firewall Access Control Dimensions (Service, Direction, User, Behavior).

Explain all of them with an appropriate example.